Deacon T. Carreon

UC Berkeley Physics and Astrophysics Major

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Aspiring scientist working towards degrees in Physics and Astrophysics. Researcher with computational and hands-on experience, at Space Sciences Laboratory, and writing experience in the Berkeley Scientific Journal. Able to grasp complex concepts, communicate clearly, and pass down knowledge to others.

Education

August 2024 - Present **University of California, Berkeley**: Physics and Astrophysics B.A.
GPA: 3.766/4.0
Coursework:
- Quantum Mechanics
- Electromagnetism and Optics
- Introduction to Statistical and Thermal Physics
- Introduction to Astrophysics

Research

Space Sciences Laboratory Fall 2025 - Present

The MAVEN Solar Energetic Particle Instrument Investigation Spring 2026 - Present

– Conducted in-depth reading about the MAVEN spacecraft, specifically its Solar Energetic Particle (SEP) instruments, and scientific objectives.

– Worked with Ali Rhamati to operate IDL software and graph variables and data from the MAVEN spacecraft.

– Analyzed spectra of Scorpius X-1, the largest extrasolar source of X-rays. Correlated MAVEN's tangent altitude from Scorpius X-1 to the X-ray particle count rate.

– Observed spectra data related to GRB 221009A, the brightest gamma-ray burst ever observed, and conducted a literature digest this subject and photon-matter interactions.

– Coded response matricies using Geant4 simulation data of gamma rays and the SEP instrument, reshaped and multiplied to SEP spectra data of GRB 221009A.

MURA Coded-Aperture Imaging With Timepix3 Fall 2025

– Worked with Aerospace Engineer Travis Curtis to analyze data from radioactive noise of onsite Americium-241 source with TimePix3 detector and Pixet software.

– Underwent training including lab safety, radiation safety, and was taught dosimetry usage, operation of pancake detectors, and handling of Americium-241.

Programming

Introduction to Experimental Physics Fall 2025 - Spring 2026

– Used the topics learned from the python decal to conduct the data analysis of my class' labs more efficiently, which also included using LaTeX for each lab report, and Jupyter Notebook for data analysis.

Introduction to Python Spring 2025

– Learned Python concepts (NumPy, Pandas, Matplotlib), the terminal, and Github. Modeled the Earth's tidal waves with physics and gravitation, with Olivia Silvia.

Other Academic Projects

Mastering Public Speaking - Mt. San Jacinto College June-July 2026

– Elective course on improving speaking in public and social settings.

Berkeley Scientific Journal Fall 2025 - Spring 2026

Interviewer Spring 2026

A Perspective on Humans on Mars, the NASA Psyche Mission, and the Pursuit of Space Exploration with Dr. Lindy Elkins-Tanton

– Interviewed UC Berkeley Space Sciences Lab Director, EPS Professor, PI of NASA Psyche Mission.

– Received key insights into how humans might explore Mars, the possibilities with exploring Psyche, and how we build public engagement with science and space exploration. Worked with Pari Sachdev.

Research and Blogs Writer Fall 2025

Planetary Defense II: Protecting Space and Earth from Our Own Satellites

– Wrote on the potential problems that can arise with the growing overabundance of satellite networks around Earth.

– Concluded that the impacts on the atmosphere from satellites completely burning up, releasing their metals and creating alumina and aerosols, is unknown.

Planetary Defense I: How Humanity is Trying to Avoid The Fate of The Dinosaurs

– Wrote on the history of asteroid impacts both within and before recorded history and the importance of planetary defense.

– Discussed missions that work to enhance planetary defense, such as NASA's and the ESA's asteroid trackers, and the 2022 NASA DART Mission.

Teaching/Mentoring

Mathematical and Physical Sciences (MPS) Scholars Mentor Fall 2025 - Present

– Mentored ten incoming students for Summer 2026.

– Mentored two first-year undergraduates during the 2025-2026 year.

Community

Golden Bear Orientation Leader 2026 August 20-24

– Co-lead a group of incoming students with Isaiah Sabiniano during the five days of freshmen orientation.

– Facilitated discussions about campus resources, safety, and methods to help excel during college.

– Supported each student, and learned about their background and interests, setting them up for success.

Cal Day 2026 April 18

– Represented the Physics department in a panel with three other MPS students to answer questions.

– Tabled with SSL, the Astronomy Department, and MPS.

SPS Shadow Day 2026 April 17

– Led admitted UC Berkeley students through my classes, research, professors and faculty.

– Participated in student panel, and helped groups get to lab tours.

MRRC Senior Weekend Hosting 2026 April 16-19

– Volunteered to host admitted UC Berkeley students at my dorm room, discussed our experiences as students.

TALKS/PRESENTATIONS

The Ambitions of a Lunar Return: NASA Artemis Program 2026 March 10

– Society of Physics Students Lightning Talk

– Why NASA is returning to the Moon, the compoments and methods it's using, the future plans, and what it means for individuals and society.

The Past, Present, and Future of Antimatter 2025 February 6

– Society of Physics Students Undergraduate Seminar

– Antimatter has always been a substance of interest in the universe, especially because of the lack of antimatter in the universe.

– Despite being mathematically plausible, our universe has come to be dominated by regular matter instead.

– Challenges with creating our own antimatter for real-world applications.

CONFERENCES/WORKSHOPS

2026 Sagan Summer Workshop - Online 2026 July 20-24

– Watched background lectures on the Roman Space Telescope, Microlensing Events, and Transiting Exoplanets.

– Partook in hands-on workshops on Google Colab for fitting single and binary lens Microlensing, and placing microlensing and transit events in galactic context.

– Conducted and presented a group project on Single Point Lens Microlensing.

248th American Astronomical Society Meeting - Pasadena, CA 2026 June 14-18

– Volunteered to hand out registration badges on 6/14, supported a large amount of the attendees.

– Attended presentations and networked with researchers working on the Roman Space Telescope, Habitable Worlds Observatory, Caltech IPAC, and more.

– Visited poster sessions of peers and other researchers, and exhibit booths of research groups to learn more about their opportunities.

Association of Science and Technology Centers 2025 - San Francisco, CA 2025 September 6-7

– Supported the session *Gardens in Space and on Earth* with Lauren Patrick, Julie Malmberg, and Jill Coons.

– Assisted in giving directions for a public science event in Yerba Buena Gardens.